7	(a)		charge exists only in discrete amounts	B1	[1]
	(b)	(i)	$E = I(R + r)$ or $V = IR$	C1	
			(total resistance =) 2.7 + 0.30 + 0.25 (= 3.25 Ω)	M1	
			$I = 9.0/(2.7 + 0.30 + 0.25)$ or $9.0/3.25 = 2.8$ A	A1	[3]
		(ii)	$V = IR_{ext}$ $= 2.77 \times 3.0$ or 2.8×3.0	C1	
			or		
			$V = E - Ir$ $= 9.0 - 2.77 \times 0.25$ or $9.0 - 2.8 \times 0.25$	(C1)	
			$V = 8.3$ (8.31) V or 8.4 V	A1	[2]
	(c)	(i)	$I = nevA$		
			$v = 2.77/(8.5 \times 10^{29} \times 1.6 \times 10^{-19} \times 2.5 \times 10^{-6})$	M1	
			$= 8.1\ (8.147) \times 10^{-6}\ \text{m s}^{-1}$ or $8.2 \times 10^{-6}\ \text{m s}^{-1}$	A1	[2]
		(ii)	*A* reduces by a factor 4 (1/4 less) or resistance <u>of Z</u> goes up by 4×	M1	
			current goes down but by <u>less than</u> a factor of 4 (as total resistance does not go up by a factor of 4) so drift speed goes up	A1	[2]